📚 DOCUMENTATION INDEX - ECOHUB QR SCANNER

Tổng hợp tất cả tài liệu hệ thống

🎯 BẮT ĐẦU TỪ ĐÂU?

👤 Bạn là End User (Người dùng cuối)?

→ Đọc: **USER_MANUAL.md**

👨‍💻 Bạn là Developer?

→ Đọc: **DEVELOPER_GUIDE.md**

🔨 Bạn cần Build app?

→ Đọc: **BUILD_GUIDE.md**

⚡ Bạn cần tra cứu nhanh?

→ Đọc: **QUICK_REFERENCE.md**

📂 TẤT CẢ TÀI LIỆU

📘 Tài liệu chính

README.md

- Quick start guide
- Features overview
- Project structure

- Tech stack
- Commands cheat sheet

DOCS.md

- Tài liệu hệ thống đầy đủ
- Tổng quan, kiến trúc, workflow
- Cài đặt development
- Cấu hình nâng cao
- Xử lý sự cố chi tiết

👤 Tài liệu User

USER_MANUAL.md

- Hướng dẫn cài đặt
- Hướng dẫn sử dụng hàng ngày
- Quy trình quay video
- Quản lý kho lưu trữ
- FAQ & troubleshooting

🖥️ Tài liệu Developer

DEVELOPER_GUIDE.md

- Setup development environment
- Code structure & modules
- Customization guide
- Testing guide
- Debugging tips
- Integration points
- Future enhancements

ARCHITECTURE.md

- System overview diagram
- Data flow diagrams
- Threading model
- State machines
- API architecture
- Performance metrics

🔨 Tài liệu Build

BUILD_GUIDE.md

- Build requirements
- Build EXE with PyInstaller
- Build Installer with Inno Setup
- Testing checklist
- Deployment guide

PYINSTALLER_FIX.md

- Fix missing static/templates
- Fix missing config.key
- Path handling for bundled app
- DEV vs PROD paths

🎥 Tài liệu Camera

CAMERA_SAFE_START.md

- Camera safe startup mechanism
- Test/Start/Stop workflow
- Camera status tracking
- Error handling
- Before/After comparison

CAMERA_SETTINGS_UPDATE.md

- Max 2 cameras change
- Camera controls moved to Settings
- Dashboard simplification
- Updated workflow

☁ Tài liệu S3

TAI_LIEU_S3_BACKEND.md

- S3 service architecture
- Upload/Download/Delete operations
- Multipart upload
- Presigned URLs
- Error handling

⚡ Quick Reference

QUICK_REFERENCE.md

- Commands cheat sheet
- Workflow summary
- Config files location
- Status indicators
- Troubleshooting quick fixes
- API endpoints
- Default values

📝 Changelog

CHANGELOG.md

- Version history
- Features added

- Bug fixes
- Performance improvements
- Breaking changes
- Migration notes

🗺️ ROADMAP ĐỌC TÀI LIỆU

Scenario 1: Lần đầu cài đặt

```
1. README.md (overview)
      ↓
2. USER_MANUAL.md (cài đặt & cấu hình)
      ↓
3. QUICK_REFERENCE.md (tra cứu nhanh)
```

Scenario 2: Build app từ source

```
1. README.md (overview)
      ↓
2. BUILD_GUIDE.md (build EXE/installer)
      ↓
3. PYINSTALLER_FIX.md (nếu gặp lỗi)
```

Scenario 3: Customize code

```
1. DEVELOPER_GUIDE.md (setup dev)
      ↓
2. ARCHITECTURE.md (hiểu kiến trúc)
      ↓
3. QUICK_REFERENCE.md (tuning parameters)
```

Scenario 4: Troubleshooting

```
1. USER_MANUAL.md → "Xử lý sự cố"
      ↓
2. DOCS.md → Section 9
      ↓
3. QUICK_REFERENCE.md → "Xử lý lỗi nhanh"
      ↓
4. Console logs
```

📊 TÀI LIỆU THEO CHỦ ĐỀ

🎥 Camera

- CAMERA_SAFE_START.md - Safe startup
- CAMERA_SETTINGS_UPDATE.md - Settings UI
- DEVELOPER_GUIDE.md → Camera Module
- ARCHITECTURE.md → Camera Flow

🤖 AI Scanner

- DEVELOPER_GUIDE.md → AI Scanner
- ARCHITECTURE.md → QR Detection Flow
- QUICK_REFERENCE.md → Tuning Parameters

📼 Recording

- DEVELOPER_GUIDE.md → Recorder
- ARCHITECTURE.md → Recording Flow
- USER_MANUAL.md → Quy trình quay video

☁️ S3 Upload

- TAI_LIEU_S3_BACKEND.md - S3 backend
- DEVELOPER_GUIDE.md → S3 Integration
- USER_MANUAL.md → Upload History
- ARCHITECTURE.md → Upload Flow

🔐 Security

- DEVELOPER_GUIDE.md → Security Considerations
- ARCHITECTURE.md → Encryption Layer
- DOCS.md → Config Encryption

🔨 Build

- BUILD_GUIDE.md - Main build guide
- PYINSTALLER_FIX.md - Build fixes
- DEVELOPER_GUIDE.md → Deployment

🔍 TÌM THÔNG TIN

Tìm theo keyword

Camera:

```
CAMERA_SAFE_START.md
CAMERA_SETTINGS_UPDATE.md
DEVELOPER_GUIDE.md → Camera Module
```

S3:

```
TAI_LIEU_S3_BACKEND.md
DEVELOPER_GUIDE.md → S3 Integration
QUICK_REFERENCE.md → S3 parameters
```

Build:

```
BUILD_GUIDE.md
PYINSTALLER_FIX.md
DEVELOPER_GUIDE.md → Deployment
```

Error:

```
USER_MANUAL.md → Xử lý sự cố
DOCS.md → Section 9
QUICK_REFERENCE.md → Quick fixes
```

API:

```
ARCHITECTURE.md → API Architecture
QUICK_REFERENCE.md → API Endpoints
DEVELOPER_GUIDE.md → Integration Points
```

Performance:

```
ARCHITECTURE.md → Performance Metrics
DEVELOPER_GUIDE.md → Performance Tuning
QUICK_REFERENCE.md → Tuning Parameters
```

📖 ĐỌC THEO THỨ TỰ (Khuyến nghị)

Cho End User:

```
1. README.md
2. USER_MANUAL.md
3. QUICK_REFERENCE.md
```

Cho Developer:

```
1. README.md
2. ARCHITECTURE.md
3. DEVELOPER_GUIDE.md
4. DOCS.md
5. BUILD_GUIDE.md
```

Cho DevOps:

```
1. BUILD_GUIDE.md
2. PYINSTALLER_FIX.md
3. DEVELOPER_GUIDE.md → Deployment
```

🆘 SUPPORT

Tài liệu không rõ?

→ Check **DOCS.md** (tài liệu đầy đủ nhất)

Lỗi khi build?

→ Check **PYINSTALLER_FIX.md**

Lỗi khi sử dụng?

→ Check **USER_MANUAL.md** → Xử lý sự cố

Cần customize?

→ Check **DEVELOPER_GUIDE.md** → Customization

📊 STATISTICS

Tổng số tài liệu: 12 files

Tổng số dòng: ~2,500+ dòng

Phủ sóng: - User Guide: 100% - Developer Guide: 100% - Build Guide: 100% - Architecture: 100% - Troubleshooting: 100%

✅ DOCUMENTATION CHECKLIST

- [x] Overview & Quick Start (README.md)
- [x] Full System Documentation (DOCS.md)
- [x] User Manual (USER_MANUAL.md)
- [x] Developer Guide (DEVELOPER_GUIDE.md)
- [x] Build Instructions (BUILD_GUIDE.md)
- [x] Architecture Diagrams (ARCHITECTURE.md)
- [x] Quick Reference (QUICK_REFERENCE.md)
- [x] Camera Guide (CAMERA_SAFE_START.md)
- [x] S3 Backend (TAI_LIEU_S3_BACKEND.md)
- [x] PyInstaller Fixes (PYINSTALLER_FIX.md)
- [x] Changelog (CHANGELOG.md)
- [x] Documentation Index (DOCUMENTATION_INDEX.md)

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